

Detecting Student Depression using Machine Learning

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Abstract—Depression is a common mental illness that is influenced by many factors, such as age, gender, sleep duration, financial status, dietary habits etc. In severe cases, it can even lead to suicide. It affects many people around the world, however, this study targets students specifically. This study aims to use machine learning (ML) to categorize student depression based on certain general and academic factors. The study involved analyzing an imbalanced dataset of 27,000 students from around the world and pursuing different degrees. Many ML models were used, such as boosting methods, Random Forest Classifier (RFC), Support Vector Classifier (SVC), Logistic Regression (LR). Before oversampling, XGBoost achieved the highest accuracy at 85% and 84.6% F1 score. After oversampling, RFC achieved the highest accuracy and F1 score at 90% using cross validation. The study found some interesting correlations between depression and some factors such as academic pressure. The study concludes that employing ML methods can help detect depression to help students before it is too late.

Index Terms—classification, prediction, depression, machine learning (ML), students

I. INTRODUCTION

Depressive disorder (also known as depression) is a common mental disorder in which people feel a persistence in low mood or a loss of pleasure or interest in activities that were previously enjoyable for extended periods of time [1]. Around 3.8% of the population suffers from depression, including 5% of adults, (4% of men, and 6% of women) [1]. Depression is 50% more common among women than among men [1]. Moreover, COVID-19 has led to a 26% increase in the prevalence of clinically elevated depression symptoms in youth, with most of the evidence being conclusive in women and mid to high income individuals [2]. Some basic characteristics of depression are the loss of interest in activities that were previously enjoyable, disrupted sleep, lower appetite and more weight loss, thoughts about suicide or dying, and academic stress in the case of students. [1], [3]. More than 700,000 people die from suicide every year, many more than this number attempted it but did not die, suicide is also the fourth leading cause of death among individuals aged 15 to 29 years [1]. The estimated number of people living with depression has increased by about 18.4% between 2005 and 2015 [4]. University students experience depression rates of 30.6%, which is substantially higher than the general population [3]. A depressed person may not realize that they

are depressed [5], this highlights the importance of utilizing modern Machine Learning (ML) models to predict depression based on certain factors. While traditional statistical methods can be still useful, ML techniques can assess a larger amount of variables for predictions, which can greatly improve efficiency and accuracy. Machine Learning prediction systems can make highly precise predictions on new data and scenarios. This study uses a generated dataset sourced from Kaggle, the dataset includes information on 27,000 students across three educational levels: high school, undergraduate, and postgraduate. This study aims to classify and predict students using Machine Learning methods, with the goal of finding the best scoring one across multiple different tested models. The main ML models used include boosting algorithms such as XGBoost (XGB), ensemble methods such as RandomForestClassifier (RFC), Support Vector Classifier (SVC) and Logistic Regression (LR). These models were employed with and without cross validation, before and after oversampling, in order to achieve the best result.

II. REVIEWING SIMILAR WORKS

Reviewing a previous work is essential and forms the basis of the research process. In this project, reviewing previous work helps us understand the causes and factors of depression and expands our understanding of depression and its causes.

Detecting depression using an ensemble classifier based on Quality of Life scales [6].

A. Objective of the Study

The study aimed to improve early detection of depression, by developing a binary classifier, This model takes advantage of knowledge in the psychological field, specifically the Quality of Life Scale (SF-20) and the Patient Health Questionnaire (PHQ-9), to predict depression based on health data.

B. Methodology

Dataset: The study used data from the National Health and Nutrition Examination Survey (NHANES), including responses from 5398 participants [6], data cleaning was applied, and missing values were handled. **Model:** An ensemble classifier combined predictions from four machine learning models: Support Vector Machines (SVM), Artificial Neural

Networks (ANN), k-Nearest Neighbors (KNN), and Decision Trees (DT), were used.

C. Key Findings

The ensemble classifier achieved a high predictive performance with: F1 Score: 0.976. Accuracy: 0.954.

D. Relevance to Current Study

This research demonstrates the potential of machine learning to accurately detect depression using health-related data. It is in line with the efforts of the current project to diagnose student depression based on a combination of psychological, physical, and lifestyle factors.

E. Conclusion

This work highlights the effectiveness of ensemble classifiers in predicting mental health. Future research should explore additional datasets and include more features and results to improve predictions.

III. METHODOLOGY

A. Dataset

1) *Dataset Acquisition:* The dataset was acquired using Kaggle, an online platform for data science and machine learning. It provides datasets, notebooks, competitions and tools for exploring the field of Artificial Intelligence (AI) and data science.

2) *Information About The Dataset:* The dataset provides synthetically generated data that simulates the real world. It includes features such as demographic information and academic information. The dataset is formatted as a CSV file where each row represents a student and each column represents a feature. The dataset contains 27,000 rows of students with 17 columns of features, and a column for the target variable. The dataset is imbalanced where 63.5% of the students in the dataset have depression, and 36.5% do not have depression.

3) Features in The Dataset:

- Gender: Gender (Male, Female).
- ID: ID to identify each student.
- Age: The age of the student.
- City: Geographic region.
- Profession: Student's current job.
- Academic Pressure: The pressure the student feels on scale from 1 to 5.
- Work Pressure: The work pressure the student feels on scale from 1 to 5.
- CGPA: Grade Point Average or other academic scores.
- Study Satisfaction: A scale from 1 to 5 on how satisfied the student is with their studying.
- Job Satisfaction: A scale from 1 to 5 on how satisfied the student is with their job.
- Sleep Duration: Average daily sleep duration.
- Dietary Habits: The student's dietary habits (Healthy, Moderate, Unhealthy).
- Degree: The student's current degree.

- "Have you ever had suicidal thoughts?": (Yes, No).
- Work/Study Hours: The total daily work and study hours.
- Financial Stress: The financial stress the student feels on a scale from 1 to 5.
- Work Pressure: The work pressure the student feels on a scale from 1 to 5.
- Family history of mental illness: (Yes, No).
- Depression: Target variable (Yes, No).

B. Data Preprocessing

This section begins with data cleaning. The dataset was relatively clean, however some cleaning was necessary to handle outliers and null values. For example, the "fillna()" function was used to fill null values in the dataset, with either the mode for categorical values or the median for continuous values. The "loc()" and "drop()" functions were used to handle out of range data, or non-meaningful data. The degrees were grouped under three main categories: High-School, Graduated, Post Graduated. Label encoding was used to transform categorical values into numerical values, without altering the inherent weight of the original values. Oversampling was used and will be discussed in a later section.

C. Exploratory Data Analysis

In this phase, the data will be looked at and visualized in order to understand the relationships between some of the features. Analysis and visualization was carried out using the pandas, matplotlib, seaborn, sklearn, and numpy libraries.

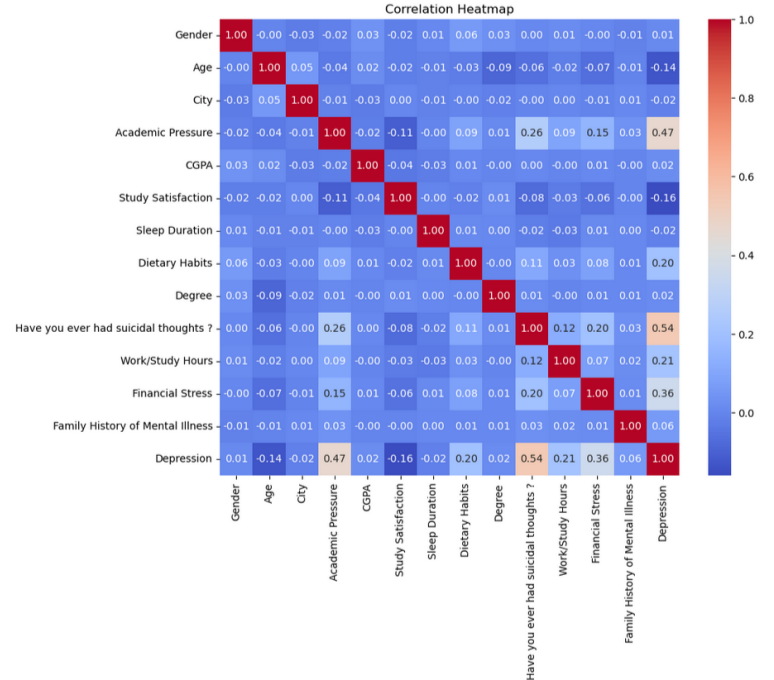


Fig. 1. Correlation Heatmap

As can be seen in Figure 1 having suicidal thoughts is the most related feature to depression, This is consistent with the fact that suicidal thoughts are a prevalent symptom of depression

[1]. Other interesting correlations can be made, such as higher academic pressure with depression, where students who feel more pressure academically are more likely to feel depressed [3]. Dietary habits are also a factor in depression, where people with depression are more likely to have unhealthy eating habits [1].

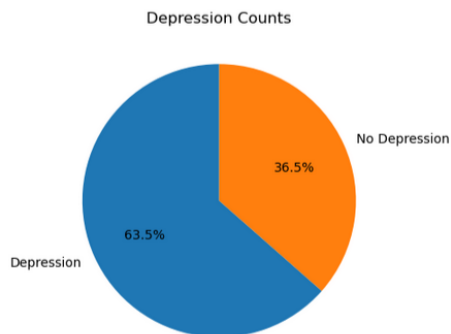


Fig. 2. Depression Pie Chart

Figure 2 shows the imbalance in the dataset as mentioned in section III-A2.

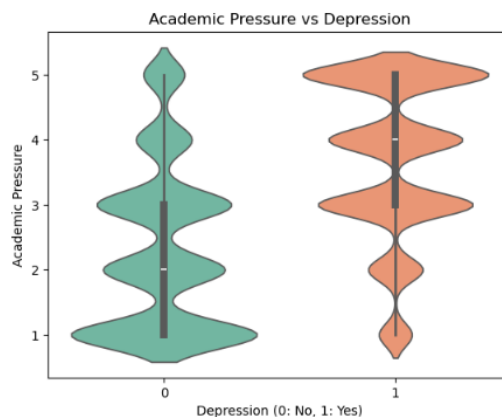


Fig. 4. (Academic Pressure vs Depression) Violin

Figure 4 shows the relationship between academic pressure and depression, illustrated by a violin plot, and it is concluded that students who had high academic stress were more frequently depressed than those who did not [3].

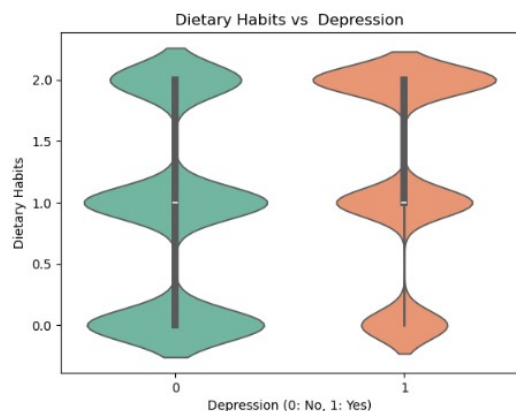


Fig. 3. (Dietary Habits vs Depression) Violin

Figure 3 shows the relationship between dietary habits and depression, illustrated by a violin plot, and it is concluded that in the category that follows a healthy diet (0), the probability of suffering from depression was lower, and it can be noted that it became higher in people who follow an average healthy diet (1), and became significantly higher in people who follow an unhealthy diet (2), this means that adhering to a good healthy diet reduces the rates of suffering from depression [1].

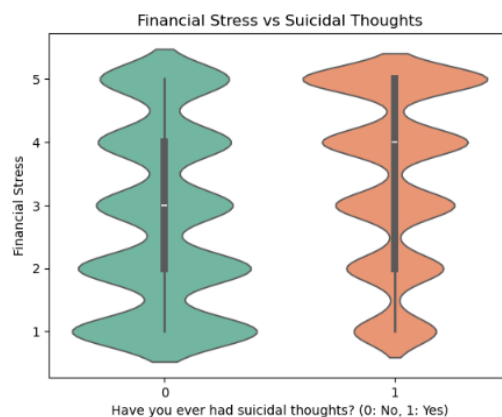


Fig. 5. (Financial Stress vs Suicidal Thoughts) Violin

Figure 5 shows the relationship between financial stress and suicidal thoughts, illustrated by a violin plot, and it is concluded that students who had high financial stress had a higher number of suicidal thoughts than those who did not have high financial stress [7].

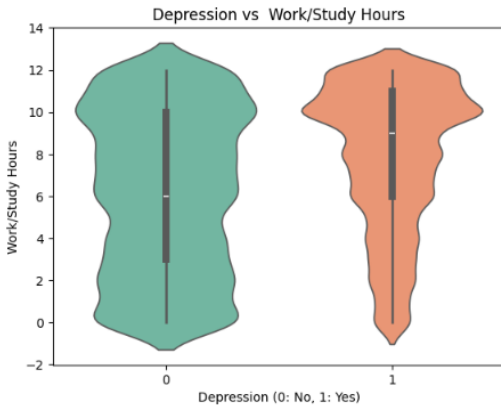


Fig. 6. (Work/Study Hours vs Depression) Violin

Examining the results of people suffering from depression shown in figure 6, it can be concluded that typically students who spend more time working and studying, are more likely to be depressed, as opposed to those who do not. This is consistent with the causes of depression [8].

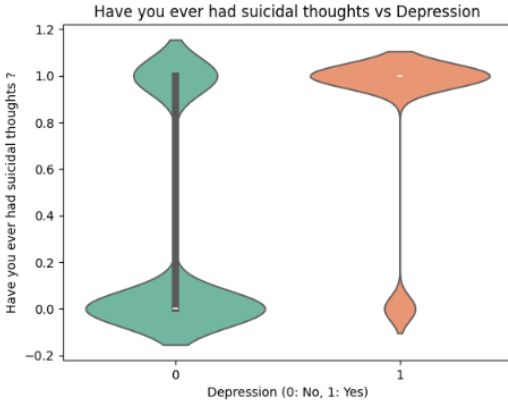


Fig. 7. ("Have you ever had suicidal thoughts?" vs Depression) Violin

It is noted from figure 7 that the people who had suicidal thoughts were, to a large extent, suffering from depression. Depression is associated with suicidal behavior, which is often translated to a long continuum of suicidal thoughts [9].

D. Machine Learning

Machine Learning is a way to predict values based on certain features. It can be used to perform complex predictions that would otherwise require a lot of time and effort [10].

1) *Random Forest*: Random Forest is a widely used ensemble learning technique for classification and regression tasks. It creates multiple decision trees during training and combines their outputs using majority vote for classification, and average of the trees for regression. Random Forest improves accuracy and helps prevent overfitting [11].

- **Ensemble Learning**: Random Forest consists of multiple decision trees each trained on a different subsample of the data, which helps control overfitting and achieves

higher accuracy. The final output is reached either by majority vote for classification tasks, or average of each tree's output for regression tasks.

- **Random Sampling**: During training, random subsets of the data are created using a method with the name of bootstrap aggregation (also known as bagging). Every tree is trained on a different subset ensuring a different training process for each tree.
- **Feature Randomness**: When splitting trees, not every feature is considered. Instead, a random subset of features for each tree is considered, which reduces correlation among the trees, improving the model's ability to generalize to different data.

Advantages of Random Forest:

- **Robustness**: Due to ensemble learning, random forest is robust to overfitting, which improves its ability to generalize.
- **Accuracy**: Random Forest provides high accuracy due to ensemble learning, making it a strong contender over other ML methods.
- **Versatility**: Random Forest supports both classification and regression tasks, increasing the fields in which it can be used.
- **Non-Parametric**: Random Forest does not make any strong assumptions about the data distribution.
- **Flexibility in Parameter Tuning**: Random Forest provides multiple parameters that can be tuned (e.g., maximum depth, number of trees) that can allow for fine tuning and depending on the dataset and task.

Disadvantages of Random Forest:

- **Interpret ability**: It can be difficult to understand how the model reached its prediction.
- **Computational Cost**: Training and prediction tasks can be computationally expensive depending on the size of the dataset.
- **Memory Usage**: Due to the amount of decision trees constructed, random forest requires a large amount of memory to store all of them. This can become problematic in very large datasets or in high dimensional data.
- **Sensitivity to Imbalanced Datasets**: Random Forest can struggle when an imbalanced dataset is present, where the majority class will dominate most of the predictions, leading to poor performance on the minority class. This calls for the use of methods such as oversampling.
- **Suboptimal Feature Importance Estimation**: Random Forest can sometimes misrecognize important features, as it tends to favor features with more levels or higher variance (such as categorical features with many variables).

Random Forest is a strong ML algorithm, due to its high accuracy and robustness to overfitting. It is also a strong contender for the dataset used, as it is relatively small which reduces its computation cost. However, the dataset is imbalanced, and as mentioned in section III-D2 this can be problematic, therefore random oversampling was used and will be discussed in a later section [11].

Random Forest

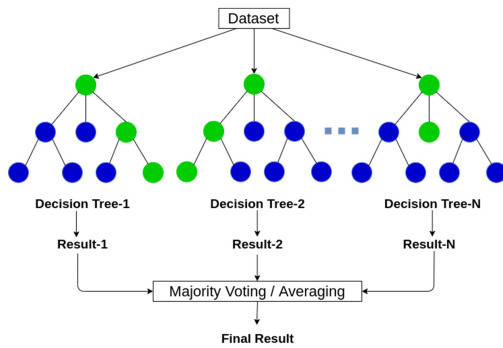


Fig. 8. Random Forest. Source: [12].

2) *Support Vector Machine (SVM)*: Support vector machine is a supervised machine learning model designed for classification and regression tasks. It works by defining the hyperplane that separates data points of different classes, in the case of classification, the optimal hyperplane is determined by maximizing the margin to reduce overfitting, without causing classification errors or maybe cause it, in the case of non perfectly separable data, the support vector machine is mathematically complex and is a computationally expensive model, designed for classification and regression tasks [13] .

- **Maximizing Margin**: Maximizing the margin between points enables the algorithm to find the best decision boundary between classes. This enables the SVM to generalize well to new data and make accurate classification predictions.
- **Kernel Trick**: The kernel trick is a method used in SVM to enable them to classify non-linear data using a linear classifier. By applying a kernel function, SVM can implicitly map input data into a higher-dimensional space where a linear separator (hyperplane) can be used to divide the classes. This mapping is computationally efficient because it avoids the direct calculation of the coordinates in this higher space.

Advantages of Support Vector Machine:

- **Effective in High-Dimensional Spaces**: SVM are highly effective when dealing with datasets that have many features (dimensions).
- **Handles Nonlinear Data with Kernels**: SVM uses kernel functions to transform data into higher dimensions, enabling the separation of classes that are not linearly separable in the original space. This flexibility is a key advantage.
- **Robust to Overfitting**: Due to the principle of margin maximization, SVMs generalize well on unseen data, reducing the risk of overfitting, especially in high-

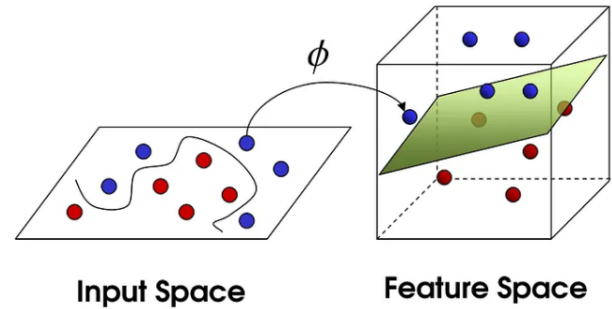


Fig. 9. Kernel Trick. Source: [14].

dimensional feature spaces.

Disadvantages of Support Vector Machine:

- **Computationally Expensive**: Training an SVM can be slow, especially for large datasets, due to the quadratic programming optimization process.
- **Requires Careful Parameter Tuning**: SVM require careful tuning of hyperparameters like the regularization parameter "C" and kernel type, which can be time-consuming.
- **sensitive to noisy data**: SVM can be sensitive to noisy data, especially if the data points are mislabeled, as they try to maximize the margin between classes.

In machine learning, support vector machines (SVMs, also support vector networks) are supervised max-margin models with associated learning algorithms that analyze data for classification and regression analysis. Developed at AT&T Bell Laboratories, SVMs are one of the most studied models, being based on statistical learning frameworks of VC theory proposed by Vapnik (1982, 1995) and Chervonenkis (1974).

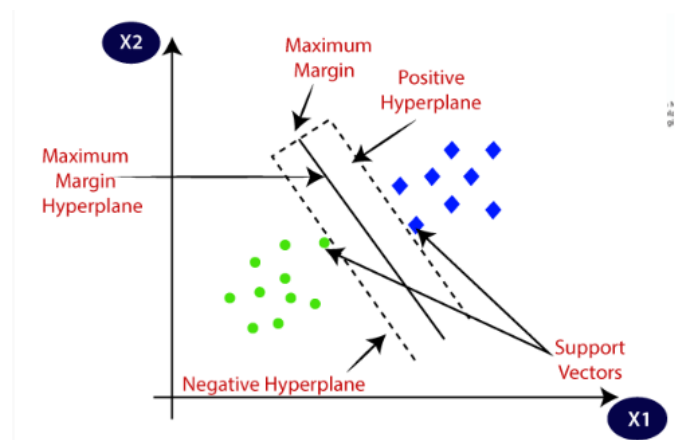


Fig. 10. Support Vector Machine. Source: [15]

3) *Boosting*: Boosting is one of the most popular machine learning techniques and can be considered a way of enhancing predictive models' accuracy by combining several weak learners' predictions. Models here are trained sequentially, each trying to correct the errors the previous one makes,

hence reducing bias. This is the reason why boosting has gained special recognition and may bring huge performance improvement for the models that are rather weak [16].

Concepts of Boosting:

- **Training Sequentially:** Training of the base model usually takes place first - the normally very basic model, such as a decision tree. This model normally gives its prediction based on the data.
- **Error Correction:** More models are then trained with the goal of correcting mistakes made by models previously trained. So, in fact, the main concept is to put more weight into samples misclassified by models from the earlier classes.
- **Combining Models:** At last, when a number of models are trained, then normally a final prediction is done by joint efforts of all of them in the form of some weighted vote, or sometimes just by averaging the votes. This is how the model ensemble will finally predict with an elevated level of precision.

Types of Boosting:

- **AdaBoost, or Adaptive Boosting,** works out to be one of the first successful boosting algorithms [16]. AdaBoost works by shifting the weights of the misclassified samples so that the next model will focus more on the problematic ones. It combines the results of weak classifiers into a single strong classifier [16].
- **Gradient Boosting:** The basic idea behind gradient boosting is to fit the model on residual errors made by previous models. The underlying idea here is the minimization of the loss function using gradient descent, hence turning out an improved model iteratively [16].
- **XGBoost:** it is an optimized and enhanced version of Gradient Boosting. All it does is bring a few techniques in parallelization, regularization, and pruning together for faster training with no loss due to its scalability advantage. XGBoost works efficiently with big datasets and finds wide applications in machine learning competitions owing to their strong performance.
- **LGBMClassifier:** LightGBM is an efficient gradient boosting algorithm from Microsoft [17]. It uses a histogram-based technique and hence is faster and more scalable than other gradient boosting models. This will help in the case of large datasets and high-dimensional problems. It uses techniques like leaf-wise growth rather than the level-wise growth concept used in other boosting algorithms; hence, it can handle complex data structures better and faster [16].

Advantages of Boosting:

- **Improved Performance:** Boosting generally improves model performance to a great degree on complex datasets.
- **Flexibility:** Boosting algorithms can be put to work on an incredibly wide range of models, especially decision trees, and for both classification and regression tasks.
- **Error Management:** Boosting reduces bias by learning from the mistakes of the previous models. Hence, it's

strong in handling complex patterns in the data.

- **Regularization:** Boosting adds techniques of L1 and L2 regularization in algorithms such as XGBoost. Thus, it avoids overfitting.

Disadvantages of Boosting:

- **Overfitting:** Boosting can easily overfit if the model is not tuned well, especially in large numbers of iterations or trees.
- **Computational complexity:** Algorithms of boosting, mainly XGBoost and LGBM, can also become computationally expensive while dealing with very big datasets.
- **Model Interpretability:** Though techniques are being developed to improve model interpretability, boosting algorithms-especially with large numbers of trees-are hard to interpret.
- **Parameter tuning:** Boosting models have a lot of hyper-parameters to be tuned for better performance; hence, the model-building process could be very time-consuming.

Boosting continues to prove to be one of the best ML algorithms available, due to its improved accuracy, being widely used across all fields and in many applications. The variety in boosting algorithms means an appropriate boosting algorithm can be used for any given task. Its high variety and accuracy makes it an excellent contender for machine learning.

4) *Bagging:* Bagging is an ensemble learning technique that reduces the variance of an estimate by bootstrapping the data set and separating it into several parts and combining its results after generating a prediction model. The model is built from multiple weaker models and aims towards higher accuracy [18].

The key aim of bagging is to boost accuracy while minimizing the chance of overfitting due to a large number of models trained.

Concepts of Bagging:

- **Constructing Bootstrap Sample:** Multiple subsets are constructed by randomly using a sampling technique with replacement of the original data set into smaller parts known as bootstrap samples. All the subsets together combine to form an original data set which means that through the rotation method, multiple data points can emerge while the others are excluded [19].
- **Training Independent Models:** A single base model or decision tree is trained to a single bootstrap sample which acts like a random generator. Each model working with its subset begins to pick patterns that are unique to its subset in the data.
- **Aggregating Predictions:** During voting, classification is the specific action that is performed in which the prediction of all models is combined to gross majority. Regression tasks typically entail making a single average out of all the model outputs
- **Acquainted Resampling Techniques:** A random forest is a special type of bagging wherein several decision trees are utilized instead of a single tree.

- Random Forest enhances the level of imperfection in the model by ensuring that only a small portion of the features are chosen for each decision tree split. It is resilient, works with overfitting, and is commonly used for classification and regression [18].
- Bagged Decision Trees: Under bagging the approach used to train the classifiers is akin to regular bagging but implemented on decision trees. These models work to minimize variance and thus are able to achieve better performance than individual simple decision trees [18].

Advantages of Bagging:

- Excessive fitting is controlled: Aggregation is helpful in reducing overfitting, as it is achieved by averaging the predictions, this is of utmost importance with high-variance models such decision trees.
- Increased Precision: It also increases the accuracy of the model predicting both the training and test datasets.
- Strength: Bagging is more robust against data as compared to the individual models since it is less susceptible to outliers and noise present in the data.
- Parallelization: Since all models are trained independently this makes it suitable for parallelization.

Disadvantages of Bagging:

- Computational Overhead: Several models being trained would increase the computational requirements especially for bigger datasets.
- Less Transparency: Since banding has ensembles as its basis it renders the model more difficult to interpret than a singular model.
- Not Always Effective: High variance models implementing bagging tend to derive substantial benefit from it, however low variance models for instance linear regression would not notice much improvement.

Overall, bagging is a technique designed to battle overfitting and prevent it by training multiple models on different subsets of the data. It captures the important parts of decision trees while reducing the problems present in them. This makes bagging an incredible algorithm for ML, and a strong contender over other ML models.

E. Oversampling

Oversampling was used in the dataset in order to balance it. Oversampling can be helpful to prevent models from becoming biased to the majority class, preventing them from accurately classifying the minority class. Random oversampling was used using the imblearn library. Random oversampling duplicates random minority class instances to balance the dataset. This proved to be useful as will be discussed in the results section.

IV. RESULTS

This section includes the results of the training, classification and oversampling phases. Model results will be looked at before and after oversampling to analyze the differences.

A. Before Oversampling

Before oversampling, most of the models results were close to each other, with some slight differences, however, xgboost will be considered as the best performing model due to its slightly better precision. Table I shows the results of the most relevant models.

TABLE I
MODEL RESULTS BEFORE OVERSAMPLING

Model	Accuracy	Recall	Precision	F1-Score
Random Forest	0.84	0.84	0.84	0.84
XGBoost	0.85	0.85	0.85	0.84
SVC	0.85	0.85	0.84	0.84
Extra Trees	0.83	0.83	0.83	0.83

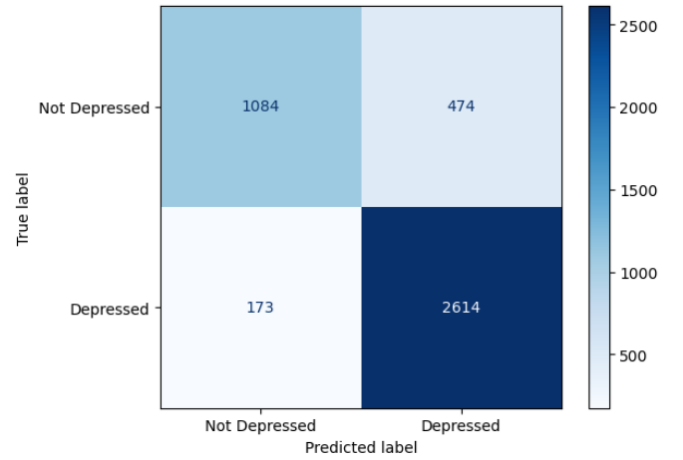


Fig. 11. Confusion Matrix of XGBoost

Figure 11 shows the predictions of the model vs the actual value. The values on the top left show the true negatives, which are the values that were predicted to be negative and are actually negative. The values on the bottom right show the true positives, which are the values that were predicted to be positive and are actually positive. The values on the top right show false positives, which are the values that were predicted to be positive but are actually negative. The values on the bottom left show false negatives, which are the values that were predicted to be negative but are actually positive. As can be seen in the figure, the model was able to correctly identify most of the cases, however, bias to the majority class can be seen, as the model seems to be detecting the students with depression better than the students without depression. As mentioned in section III-E, having an imbalance in the dataset will naturally bias the model towards the majority class, which is students with depression.

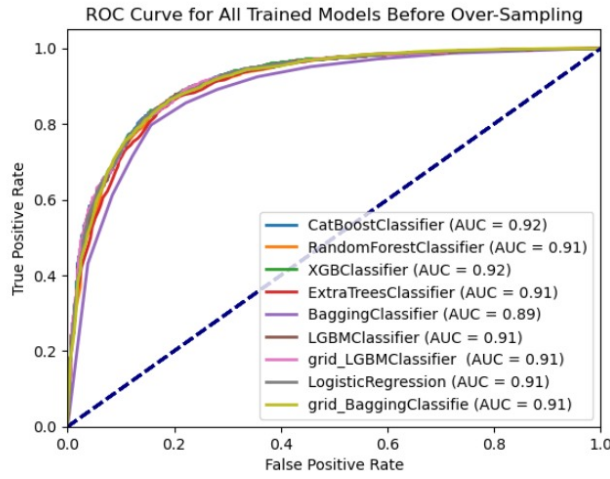


Fig. 12. ROC Curve before Oversampling

Figure 12 represents the Receiver Operating Characteristic (ROC) curve of the models before oversampling was performed. The ROC curve is a graphical representation used to evaluate the performance of a binary classifier. It shows the trade-off between the True Positive Rate and the False Positive Rate at various classification thresholds. The Area Under The Curve (AUC) shows the overall ability of the classifier to distinguish between the positive and negative classes, The higher the AUC, the better the model is. As can be seen in figure 12, XGBoost achieved the highest AUC. This aligns with the previously discussed results.

B. After Oversampling

After oversampling, random forest and extra trees obtained excellent results compared to the rest of the classifiers, however, Extra Trees performed the best in all metrics, Table II shows the results of the most relevant models.

TABLE II
MODEL RESULTS AFTER OVERSAMPLING

Model	Accuracy	Recall	Precision	F1-Score
CatBoost	0.84	0.84	0.84	0.84
Random Forest	0.90	0.90	0.90	0.90
XGBoost	0.84	0.84	0.84	0.84
SVC	0.83	0.83	0.83	0.83
Extra Trees	0.91	0.91	0.91	0.91

Figure 13 shows the predictions of the model vs the actual value. The confusion matrix is the same as the one explained in the above section. Extra Trees did not achieve a significant result before oversampling, however, after oversampling it was able to achieve the best result. This proves the significance of oversampling as random forest and extra trees were able to do much better after it was performed. Figure 13 shows the confusion matrix for extra trees classifier, as it was the best performing model.

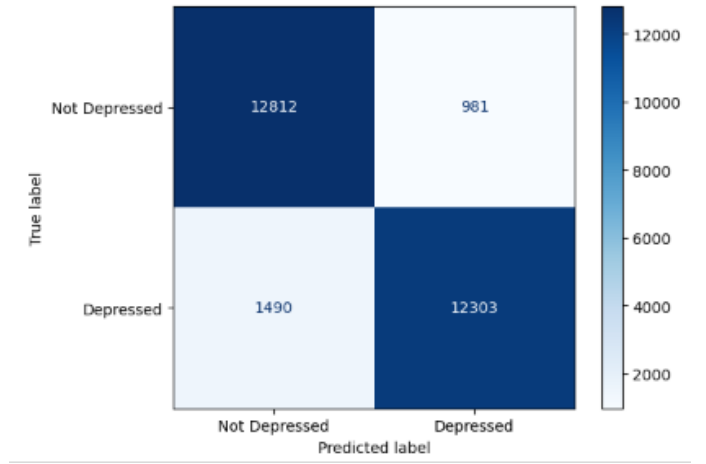


Fig. 13. Confusion Matrix of Extra Trees

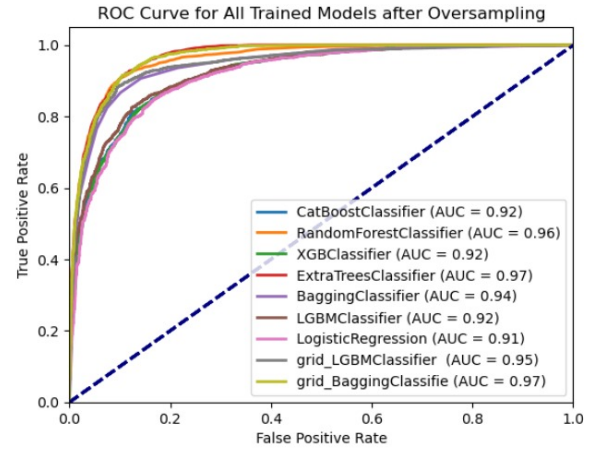


Fig. 14. ROC Curve after Oversampling

Figure 14 represents the ROC curve after oversampling. As can be seen, most of the models have shown improvements, particularly the extra trees, bagging and random forest classifiers. This is consistent with the previous results and proves the importance of oversampling overall, and especially in bagging style models.

V. CONCLUSION

The paper evaluated multiple ML models using supervised machine learning, to predict student depression based on multiple factors academic and non-academic. According to the results, XGBoost achieved the best results prior to oversampling, with an 85% accuracy and 84% f1-score. After oversampling, extra trees achieved the best results with 91% accuracy and 91% f1-score. This model can be integrated into university systems to predict depression for each student and help them with it. In addition, the study was able to find the factors most correlated to depression, which can be useful to detect early signs of depression, such as suicidal thoughts, or high academic pressure. Universities can use these findings to help address these issues at an early time, and stop them

from reaching dangerous levels. This can also improve the well-being of university students and the general health of students by implementing these ML models to enhance the health programs in universities.

Significant improvements have been made in this field, however more can still be done. Not enough information about depression is gathered from students, which can limit the creation of accurate models about this topic. Moving forward, research could improve analysis of depression, gather more information from students and perform better statistical analysis to improve detection of depression.

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